

Ligninger

Øvelse 1 Løs følgende ligninger.

1) $3x + 5 = 14$

2) $2x - 7 = 3$

3) $5x + 2 = 17$

4) $4x - 9 = 11$

5) $6x + 1 = 25$

6) $8x - 3 = 13$

7) $x + 12 = 5$

8) $3x = 21$

9) $2x + 10 = 4$

10) $7x - 4 = 17$

Øvelse 2 Løs følgende ligninger.

1) $2(x + 3) = 10$

2) $3(x - 2) = 9$

3) $4(2x + 1) = 20$

4) $5(x - 4) = 0$

5) $2(3x + 2) = 16$

6) $3x + 2 = x + 8$

7) $5x - 3 = 2x + 9$

8) $4x + 1 = 2x + 7$

9) $6x - 5 = 3x + 4$

10) $2x + 11 = 5x - 4$

11) $7x - 2 = 4x + 10$

12) $3x + 8 = 6x - 1$

13) $9x - 7 = 5x + 9$

14) $2(x + 4) = 3x - 1$

15) $4(x - 2) = 2x + 6$

16) $3(x + 5) = 2(x + 8)$

- 17) $4(x - 1) = 2(x + 3)$
- 18) $5(2x - 3) = 3(x + 4)$
- 19) $2(x + 6) - 4 = 3(x - 1)$
- 20) $3(2x - 1) + 5 = 4(x + 2)$

Øvelse 3 Løs følgende ligninger.

- 1) $\frac{x}{2} + 3 = 7$
- 2) $\frac{x}{3} - 1 = 4$
- 3) $\frac{2x}{5} + 1 = 5$
- 4) $\frac{x}{4} + 2 = 5$
- 5) $\frac{x + 1}{2} = 4$
- 6) $\frac{x - 3}{3} = 2$
- 7) $\frac{2x + 1}{3} = 3$
- 8) $\frac{3x - 2}{4} = 1$
- 9) $\frac{x}{2} + \frac{x}{3} = 5$
- 10) $\frac{x}{4} - \frac{x}{6} = 1$

Øvelse 4 Isolér x i følgende ligninger.

- 1) $ax + b = c$
- 2) $2x + a = b$
- 3) $ax - 3 = c$
- 4) $bx + a = 2c$
- 5) $a(x + 1) = b$
- 6) $2(x - b) = c$
- 7) $ax + b = cx + d$
- 8) $a(x + b) = c(x + d)$

Øvelse 5 Løs følgende ligninger.

$$1) \frac{2x+3}{5} = \frac{x-1}{2}$$

$$2) \frac{3x-1}{4} = \frac{x+5}{3}$$

$$3) \frac{x+2}{3} + \frac{x-1}{2} = 3$$

$$4) \frac{2x-1}{3} - \frac{x+2}{4} = 1$$

$$5) \frac{4x+1}{6} = \frac{2x-3}{4}$$

$$6) \frac{x}{2} - \frac{x-1}{3} = \frac{5}{6}$$